Shiv Nadar University - Chennai

Department of Computer Science

25CS2741 Fundamentals of Internet of Things + Lab

Date: 24/08/2026 (Week 6) **Continuous Lab Evaluation-7** II Yr/ 3rd Sem B. Tech CSE(IoT)A&B

Allotted Marks: 10 (Hardware Implementation + Code Efficiency + Viva-Voce)

Topics: 8051 Tool Chain setup – Development Board – AT89S52

Strictly No to using **Arduino APIs, HAL libraries, RTOS services!**

Supporting Documents: **1.** Board Schematics **2. HD44780 Datasheet**

Hardware Debug: Multimeter, DSO

16×2 LCD Driver Development and Dynamic Display Control

Problem Statement:

Develop an Embedded C LCD driver for the AT89S52 that initializes a 16×2 character LCD, positions the cursor, displays characters and strings, updates values, and clears the display.

Task 7A — LCD Initialization and Character/String Display

Develop a 4-bit GPIO-based LCD driver by deriving the required command sequence from the HD44780-compatible LCD datasheet.

- Configure the LCD in 4-bit, 2-line mode.
- Implement *lcd_init(), lcd_cmd()* and *lcd_data()*.
- Display individual characters and strings.
- Display text on both rows.

Task 7B — Cursor Positioning and Dynamic Value Update

Develop LCD cursor-positioning and data-update functions.

Implement:

lcd_goto(row, column);

lcd_puts(string);

lcd_print_number(value);

Task 7C — 4×4 Keypad-Based Character Movement

Develop an interactive LCD application in which a character is moved across the 16×2 display using the 4×4 keypad.

Assign four keys as directional controls:

Key	Action
2	↑ Move Up
8	↓ Move Down
4	← Move Left
6	→ Move Right

Hints to develop :-

1. The standard 16×2 HD44780-compatible LCD is normally a parallel interface.

It typically has:

- RS
- RW
- E
- D4–D7 in 4-bit mode or D0–D7 in 8-bit mode
- VCC, GND, contrast (V0)

2. What does "LCD initialization" means ?

Think of it as: 8051 | | commands ▼ HD44780 LCD Controller | ├── Interface mode ├── Display configuration ├── Cursor configuration ├── Display ON/OFF └── Clear/position	**16 characters per row** **×** **2 rows** • The LCD controller maintains an internal character-address memory called DDRAM. The first line and second line correspond to different DDRAM address ranges. • **Discover this from the LCD datasheet.**

LCD initialization = sending a defined sequence of commands that places the LCD controller into the required operating mode.

3. **How to set the character size?**
   With a normal HD44780 16×2 LCD, you generally don't program the font size like on a graphical LCD. The character font is built into the LCD controller.
4. **How Does the LCD Receive Information?**
   - **Which pins are used for transferring data and which pins control what the data means?**
5. **The 8051 needs to control:** RS, E, D4-D7
6. **What is RS?**
7. **What is the Enable Pin?**
8. **Why 4-Bit Mode?**

LCD Initialization

Develop an ***lcd_ init()*** function that configures the LCD for 4-bit, 2-line operation.

Find the relevant commands in the HD44780-compatible LCD datasheet.

Function Set	**Display Control**	**Entry Mode**	**Clear Display**
4-bit 2-line 5×8 font	Display ON Cursor ON/OFF Blink ON/OFF	Cursor moves left/right Display shift/no shift	Clear DDRAM Return cursor to home.

Initialize the LCD and display the character A at the first position of the first row.

Ask them:

1. How do you initialize the LCD?
2. How do you select the first position?
3. How do you tell the LCD that A is data?
4. How do you transfer the character?

They should eventually develop functions like:

lcd_init();

lcd_cmd(...);

lcd_data('A');

Important Outcome :

How to read a peripheral datasheet, identify command fields, derive register/command values, implement a low-level driver, and experimentally validate the hardware.